n8n RSS Summarizer: Complete Deployment Guide

System Version: n8n 2.44.6 • **Environment:** Docker Container • **Integration:** OpenRouter & Google Sheets

> **Executive Summary:** This document describes the architecture, Docker deployment, configuration, and execution of the automated RSS feed summarizer that extracts articles from BBC News Technology, generates 2-sentence plain-English AI summaries using OpenRouter, and records them in Google Sheets.

1. Workflow Architecture

The workflow consists of 6 interconnected nodes arranged sequentially with dedicated AI model routing:

- **Manual Trigger:** Initiates on-demand workflow runs (or Schedule Trigger daily at 08:00 in scheduled variant).
- **RSS Read:** Queries RSS feed endpoint (`https://feeds.bbci.co.uk/news/technology/rss.xml`).
- **Limit Node:** Slices feed to the top 5 newest articles.
- **Basic LLM Chain:** Uses expression templating for 2-sentence summarization with automated retries (3 tries, 2000ms delay, continue on fail).
- **OpenRouter Chat Model:** Powered by `openrouter/free` routing to active free models.
- **Google Sheets Node:** Appends formatted rows (`Title` , `Link` , `Published` , `Summary`) into the spreadsheet.

2. Docker Deployment

To run the n8n environment in Docker with data persistence:

Command Line (Docker CLI)

```
docker run -d \
 --name n8n \
 -p 5678:5678 \
 -e TZ=Asia/Kolkata \
 -e GENERIC_TIMEZONE=Asia/Kolkata \
 -v n8n-class_n8n_data:/home/node/.n8n \
 docker.n8n.io/n8nio/n8n:latest
```

Docker Compose

```
services:
  n8n:
    image: docker.n8n.io/n8nio/n8n:latest
    container_name: n8n
    restart: unless-stopped
    ports:
      - "5678:5678"
    environment:
      - TZ=Asia/Kolkata
      - GENERIC_TIMEZONE=Asia/Kolkata
    volumes:
      - n8n-class_n8n_data:/home/node/.n8n
```

3. Importing and Running the Workflow

```
 # Import the workflow into the n8n container
./scripts/import-workflow.sh

# Run end-to-end execution headless from terminal
docker exec -e N8N_RUNNERS_BROKER_PORT=5689 n8n n8n execute --id=jSTdUWRulww8JA2H
```

4. Production Troubleshooting

Issue	Root Cause	Resolution
Error 403: access_denied	User email not listed under Test Users in Google Cloud OAuth consent screen.	Add developer/tester Google account under Google Cloud Console → OAuth consent screen → Test users.
Service Disabled (Drive/Sheets)	Google Sheets or Drive API not enabled on Google Cloud project.	Activate both APIs in Google Cloud Console API Library.
Missing Document Parameter	Spreadsheet not selected or empty ID.	Select the spreadsheet from dropdown or pass spreadsheet URL / ID into documentId parameter.
columns.schema required	n8n version 4.4+ requires column schema metadata.	Include schema definition array specifying Title, Link, Published, and Summary columns.